

Optimization of Area and Dynamic Range for an Intrinsic gain stage

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Abstract—In this report an approach of optimizing the area and dynamic range for an intrinsic gain stage will be presented. Those specifications do not have a common optimum; therefore, a figure of merit will also be analysed.

I. INTRODUCTION

The approach was based in the gm/ID methodology in which the transconductance efficiency is used as a proxy, since it is a width-independent figure of merit. By setting a specific inversion region of the device we can derive important insights, such as the area of the device, the speed, the dynamic range or the gain. The schematic that was given to optimize is presented below.

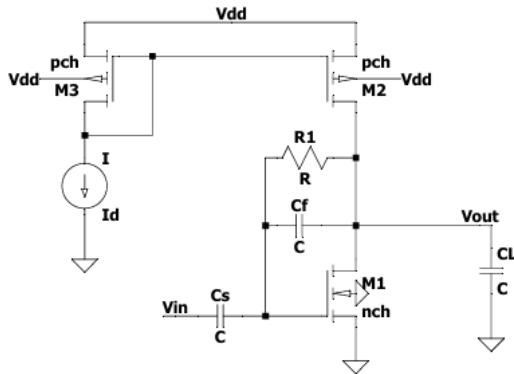


Fig. 1. Schematic of the Intrinsic gain stage.

The objective is to minimize the area of M1 and M2 while maintaining a large dynamic range. By observing the given schematic, there were already some intuition in where the devices should be biased approximately. In order to maximize the dynamic range, we expect the M1 to be biased in weak inversion while the M2 in strong inversion. By biasing the M1 in weak inversion its channel noise is expected to increase but at the same time the amplification of the signal also increases.

The transconductance of M1 is directly related to the open loop gain of the system, and therefore increasing its transconductance, improves the dynamic range. Moreover, the V_{dsat} of the device decreases, allowing a larger voltage swing.

The transistor M2 is preferable to be biased in strong inversion, since it will generate less noise. Furthermore, its transconductance does not contribute in the amplification thus it should be minimized. However, the upper limit is that the more it operates in strong inversion the more its V_{dsat} increases limiting the voltage swing. Regarding the area, the ideal case would be to minimize the transistor's size. However, the static error and the settling time, which define the required open loop gain and the bandwidth, set a limit on how small the device can be, since their dimensions directly affect their transconductance and their intrinsic resistance.

II. OPTIMIZATION APPROACH

After examining the dependency of the parameters we decided to sweep only two variables, the transconductance efficiency of the M2 transistor and the capacitance ratio (CR) between the gate-source parasitic capacitance of the gain device and the feedback capacitances. Moreover, a self-loading loop is integrated in the two loops, which update the variables taken into account the parasitic of the devices.

In the approach, it is assumed that the intrinsic gain of the M1 and M2 devices are similar, in order to find the length of the devices through interpolation. The validity of this assumption will be checked by the convergence between the modelled and the simulated results. The flowchart of the code is presented below 2.

The outer loop sweeps the values of CR, while the inner loop the values of gm/ID_p . In every CR iteration the beta is being updated.

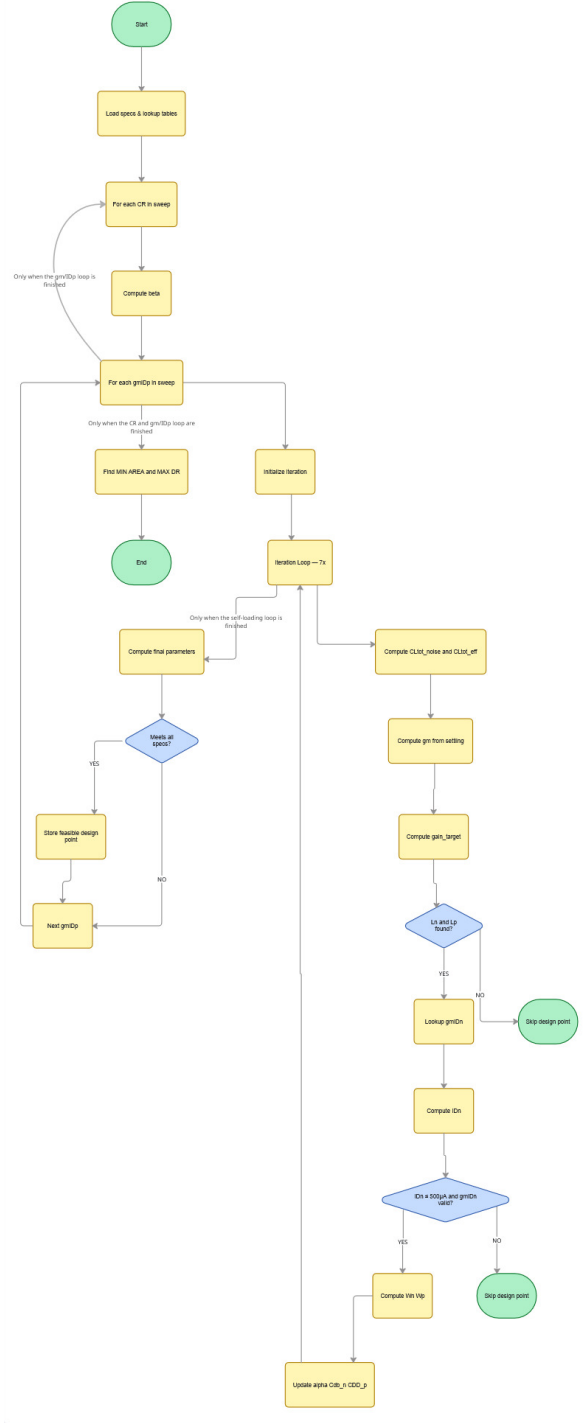


Fig. 2. Code Flowchart.

$$\beta = \frac{1}{(1 + G)(1 + CR)} \quad (1)$$

In the inner loop, the self-loading values are being initialized and gm/ID_n is set to 10 for the calculation of the first length of the NMOS. Five iterations were chosen to take into account self-loading. It is worth mentioning that since gm/ID_n is updating within the

self-loading iterations, its value is not locked to 10.

The objective in the self-loading loop is to extract the gm of the gain device and the CL_{tot} considering the parasitic capacitances of the M1 and M2 transistors. This will ensure that the calculations of the static error and the settling time are accurate. Moreover, we decided to size the transistors inside the self-loading loop because the method we used to find their lengths and widths utilize the gm/ID and the gm of the devices which unavoidably are affected by self-loading.

Delving deeper into the self-loading loop, at first the total load capacitance was declared from the noise constraint as discussed in the lecture.

$$C_{Ltot} = \frac{\alpha k_B T}{\beta \cdot V_n^2} + C_{dbn} + C_{ddp} \quad (2)$$

This is the most important part of the loop, since here the parasitic are inserted, and they affect any other variable such as the α (it is the noise excess factor), the gm , the current and hence even the dimensions of the devices. The reason we chose to add C_{dbn} and not C_{ddn} , which corresponds to the total parasitics at the drain of the NMOS is because the theory slides follow the same approach; however, for the PMOS we add all its parasitics at its drain.

The gm is calculated from the bandwidth, C_s from the closed loop gain and C_{gs} from the capacitance ratio, CR .

$$C_{Ftot} = \frac{C_{Ltot}}{FO \cdot G + 1 - \beta} \quad (3)$$

$$C_s = G \cdot C_{Ftot} \quad (4)$$

$$C_{gs} = CR \cdot (C_s + C_{Ftot}) \quad (5)$$

In this point, we calculated the gain target of the interpolation function in order to find the length of the devices. We had to match the intrinsic gain of the devices with a gain target considering that the open loop gain is:

$$open\ loop\ gain = \frac{gm}{gds_n + gds_p} = \frac{gm}{gds_n} \frac{1}{1 + \frac{gds_p}{gds_n}} \Rightarrow$$

$$\frac{L_O}{\beta} (1 + \frac{gds_p}{gds_n}) = \frac{gm}{gds_1} \Rightarrow \frac{gm}{gds_n} = \left(1 + \frac{\frac{gm}{gds_n}}{\frac{gm_p}{gds_p}} \right) \frac{L_O}{\beta} \Rightarrow$$

$$\frac{gm}{gds_n} = \frac{gm}{gds_p} = \frac{L_O}{\beta} (1 + \frac{gm_p}{ID_p})$$

By assuming that the intrinsic gain of the devices are approximately similar, we can have the following

targeted gain for the devices.

$$Gain_{targeted} = \frac{L_O}{\beta} \left(1 + \frac{g_{mp}}{I_{Dn}} \right) \quad (6)$$

At this moment, we state that after each interpolation or lookup function that returns a scalar we have inserted an if statement. The reason is that these functions can return NaN values which should be excluded from our design points. If a NaN values is encountered, the compiler goes out of the self-loading loop and the first command after its exit is an if statement, checking if there are any NaN values. If there are, it breaks the current loop of gm/IDp. For this reason at the beginning of the gm/IDp loop we empty our variables so we are sure that there are not any previous design parameter saved and mess our design points.

We continue to find the updated values of gm/IDn, IDn, and the current densities of the devices so we can calculate the widths as described in the lecture. The next step, is to calculate gmp from the gmp/IDp and the current we just found, in order to update the excess noise factor with the self-loading (this will happen from the second iteration at first we initialized the capacitances at zero as we discussed, both gamma values have been assumed 2/3).

$$\alpha = \gamma_n \left(1 + \frac{\gamma_p g_{mp}}{\gamma_n g_m} \right) \quad (7)$$

Following the self-loading loop, we calculate the actual C_L and C_F that we will use in our LTspice schematic by subtracting the parasitics we added at the beginning. At the end of the loop we extract the parasitic capacitances of the devices utilizing the lookup function.

Out of the self-loading loop, after checking if we had any NaN value we calculate the exact settling time from the bandwidth and the given dynamic error.

$$t_{s,act} = \frac{\ln(1/\epsilon_d)}{\omega_{u,act}} = \frac{\ln(1/\epsilon_d)}{\frac{\beta \cdot g_m}{C_{Ltot}}} \quad (8)$$

Afterwards, we extract the intrinsic gain of the devices through the lookup function, in order to find the intrinsic resistance of the devices. By calculating the loop gain, we can find the actual static error.

$$se = s_e = \frac{1}{1 + L_{0,act}} = \frac{1}{1 + \beta \cdot g_m \cdot \frac{r_{dsn} \cdot r_{dsp}}{r_{dsn} + r_{dsp}}} \quad (9)$$

In the next step, we calculate the noise, the dynamic range and the area.

$$V_{noise,rms} = \sqrt{\frac{\alpha k_B T}{\beta C_{Ltot}}} \quad (10)$$

$$DR = 20 \log_{10} \left(\frac{\frac{V_{dd} - V_{DSAT,p} - V_{DSAT,n}}{2\sqrt{2}}}{V_{noise,rms}} \right) \quad (11)$$

$$se = s_e = \frac{1}{1 + L_{0,act}} = \frac{1}{1 + \beta \cdot g_m \cdot \frac{r_{dsn} \cdot r_{dsp}}{r_{dsn} + r_{dsp}}} \quad (12)$$

In the last part of the loop, an if statement checks if the design points that have been selected until now violate our specifications. Outside the loops we print and plot our results.

III. COMPARISON OF MODELLED AND SIMULATED RESULTS

At first, the comparison matrix is presented in table I. Moreover, the plot presenting the static error and dynamic error from the LTspice and the modelled version in Matlab are presented in figures 3 and 4 respectively. Finally, the voltage noise power spectral density is plotted from the LTspice in figure 5.

TABLE I
COMPARISON BETWEEN THE MODELLED RESULTS AND THE SIMULATED ONES.

Parameter	Model	SPICE	Error (%)
W_n [μm]	29.00	29.00	-
L_n [μm]	0.1976	0.1976	-
W_p [μm]	11.99	11.99	-
L_p [μm]	0.5325	0.5325	-
Total Area [μm^2]	12.12	12.12	-
CR	0.0314	0.0316	≈ 0.10
DR [dB]	69.78	76.29	≈ 8.53
$(g_m/I_D)_n$ [S/A]	13.75	13.61	≈ 1.02
$(g_m/I_D)_p$ [S/A]	2.26	2.22	≈ 1.77
Output Noise [Vrms]	88.08	94.253	≈ 6.55
Settling Time [ns]	5.2	5.53	≈ 5.97
Static Error [%]	7.59	8.51	≈ 10.81
I_D [μA]	383.22	382	≈ 0.32

There are several sources that may lead to the deviation of the results between the modelled version and the LTspice. Firstly, in the model we account only the noise of M1 and M2, neglecting the contribution of M3, which causes a slight underestimate of the total output noise. Another reason could be that during the use of the lookup function, we don't specify the actual VDS of the transistor which may contribute to slight deviations due to short channel effects, such as channel

length modulation, and DIBL. Finally, we assumed a gamma of 2/3 and we did not take into account the flicker noise.

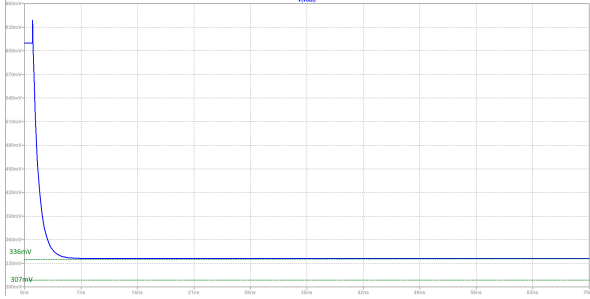


Fig. 3. Transient analysis with the static error.

In figure 3, the voltage step is delayed for 1ns in order to make visible the peak when the voltage step is applied to the amplifier. This is caused by the capacitive feedthrough of the loop capacitors. The amplifier does not respond immediately, thus the increasing voltage at the input node forces the voltage at the output to increase. When the amplifier responds we observe the expected behavior.

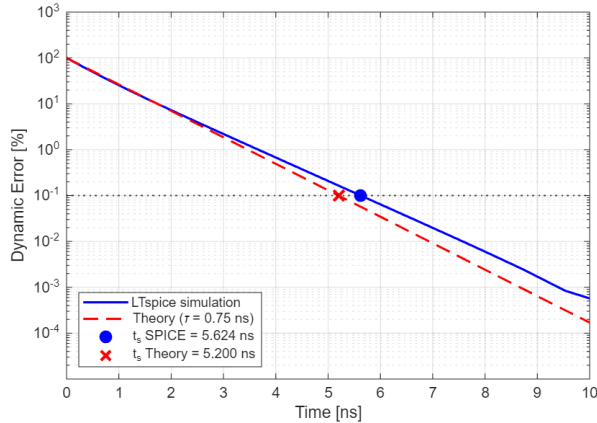


Fig. 4. Dynamic error.

In Fig. 4, the theoretical curve closely matches the LTspice simulation. The small deviation can be attributed to non-linearities in the BSIM transistor model and to simplifications in the first-order dynamic error expression. The slightly different slope between the two curves indicates a small discrepancy in the effective bandwidth.

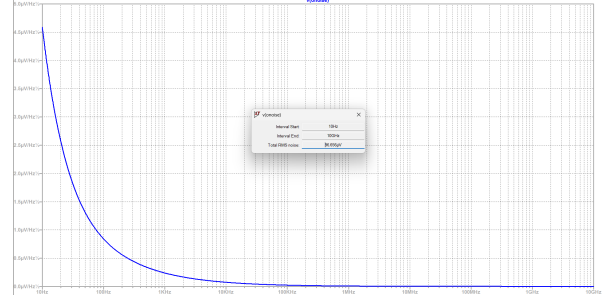


Fig. 5. Noise voltage spectral density.

IV. DECIDING AN OPTIMUM DESIGN

The decision of an optimum design point, requires a compromise between minimum area and dynamic range since they are not independent variables. As it is analysed in the introduction, different inversion regions lead to optimally the minimum area and the maximum dynamic range. We introduce a Figure of Merit in order to find an optimum between them.

$$FoM = \frac{DR}{Area_{min}} \quad (13)$$

The higher the FoM value is, the best compromise we achieve. In figure 8, we can see that the maximum FOM is very close to the minimum area. This is reasonable, since we can see in figure 6 and 7 that the DR grows slower with CR compared to the area (note that DR is plotted in db but still grows slower than the area with CR). Moreover, we can observe that the optimum is not limited by the sweep range. This is becomes apparent from the fact that the optimum is achieved for a $gm/idp = 2.3$ which is not the smallest in our sweep.

Regarding the behavior of the graphs, by increasing CR the area increases since the Cgs of the gain device increases. The bigger the gain device is the more its transconductance, and hence the higher the dynamic range.

Furthermore, We observe a dip in the FOM at very low CR, which is caused by the sudden increase of the area at low CR. The reason that happens is due to the gain requirements, and hence due to the transconductance requirements. Eventually, the device cannot become infinitely small, because its transconductance will be limited; therefore, its transconductance efficiency improves by getting bigger size at small really small CR.

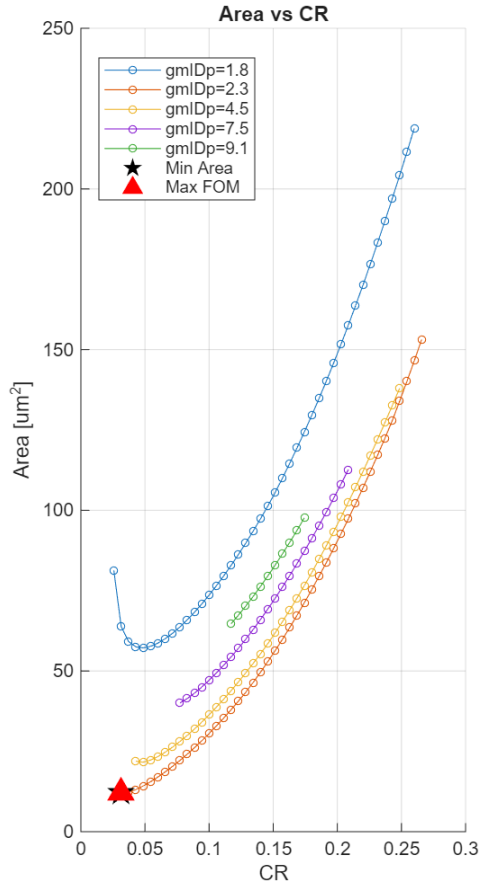


Fig. 6. Area over the capacitance ration.

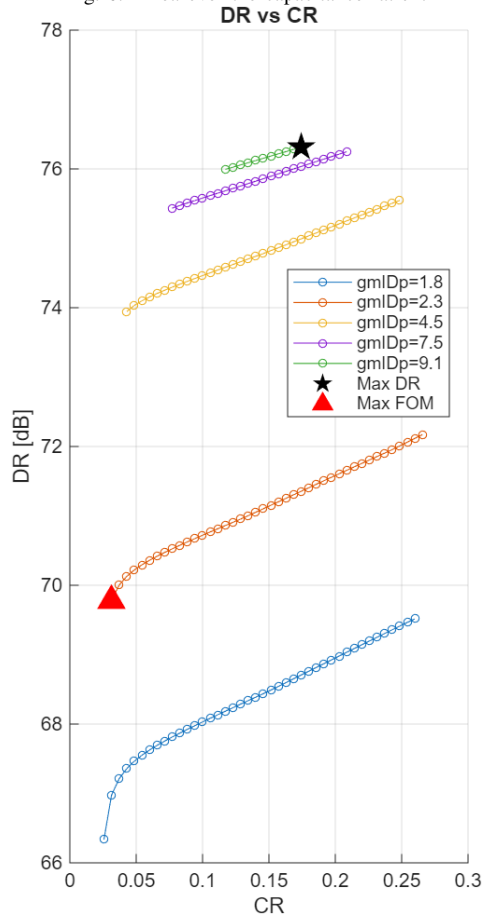


Fig. 7. Dynamic range over the capacitance ratio.

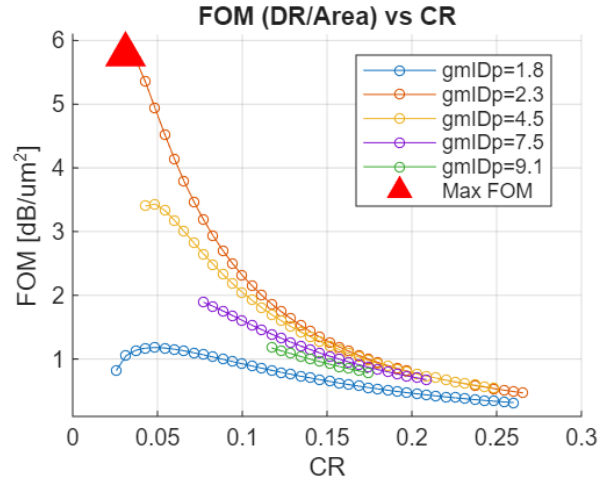


Fig. 8. Figure of Merit.